

Sparse-Latent Koopman Autoencoders for Multibasin Dynamical Systems

Anonymous Authors

Abstract

Koopman autoencoders learn coordinates in which nonlinear dynamics are advanced by a linear latent transition. Multibasin systems make this goal structurally difficult: local linearizations, and sometimes linearizations valid across individual basins, can exist around individual attracting sets, but different basins generally require incompatible finite-dimensional, state-reconstructing linear descriptions. We study sparse-latent Koopman autoencoders, where the encoder is biased to activate only a small subset of latent coordinates. The central hypothesis is that this active support can make the relevant basin or local linear law inspectable for the current state. We test this hypothesis by separating forecasting from interpretability, asking whether supports agree with basin labels in controlled benchmarks, whether they can route local predictors without oracle basin labels, and whether sparse-latent models remain competitive at long horizons. The evidence supports a qualified view. On states far from basin boundaries, sparse representations often form stable supports whose members come from a single basin, suggesting that the encoder recovers part of the attractor organization. However, exact support identity is not a reliable global invariant; it is sensitive to thresholds, basin boundaries, and equivalent latent coordinate choices. For deployment, fixed-size support routing is more robust than thresholded exact matching, and support-conditioned local predictors often improve on applying one global latent transition everywhere. Long-horizon forecasting results show that sparse-latent models remain competitive beyond controlled multibasin benchmarks, with stronger transition structure becoming most useful at the longest horizons. The main conclusion is therefore not that each basin has one exact support, but that induced latent sparsity can reveal support families that help identify the relevant basin or local predictor.

1 Introduction

Learning accurate and interpretable models of nonlinear dynamical systems remains a central problem in machine learning, scientific computing, and control. One particularly attractive direction is to seek a representation in which the nonlinear dynamics can become locally linear, since linear models support efficient prediction, estimation, and control while remaining amenable to classical solving tools such as linear quadratic regulators (LQR) (Kalman, 1960) and model predictive control (MPC) (Mayne et al., 2000; Grüne and Pannek, 2017). Classical linearization theory gives local topological conjugacies near hyperbolic equilibria (Grobman, 1959; Hartman, 1960), and Koopman eigenfunction theory extends related linearizing coordinates to entire basins of attraction for stable equilibria and periodic orbits (Lan and Mezić, 2013; Kvalheim and Revzen, 2021). So indeed, a bridge between nonlinear and linear dynamics can be found theoretically.

Koopman operator theory provides a natural way to do this. Instead of evolving the state in the nonlinear dynamical system directly, Koopman theory studies the evolution of observables under a linear operator acting on a lifted function space (Koopman, 1931; Koopman and Neumann, 1932; Mezić, 2005; Brunton et al., 2022). This viewpoint has inspired a large line of literature on data-driven approximations of Koopman dynamics, such as DMD and eDMD (Schmid, 2010;

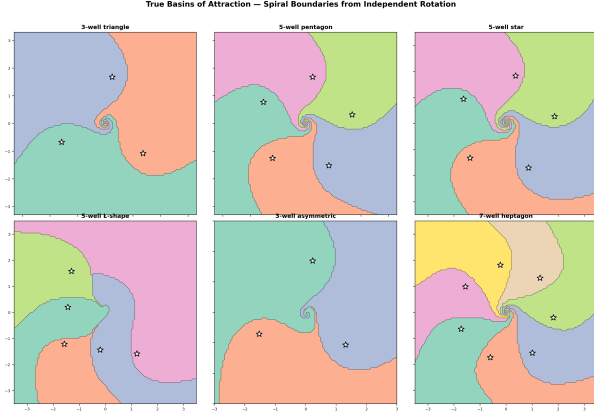
Williams et al., 2015, 2016), and applications of these linear predictors for nonlinear control via MPC (Korda and Mezić, 2018) or LQR (Mamakoukas et al., 2019). More recently, Koopman autoencoders which learn this lift from data have emerged from integrating deep learning with Koopman theory (Takeishi et al., 2017; Lusch et al., 2018; Otto and Rowley, 2019; Azencot et al., 2020; Shi and Meng, 2022; Fathi et al., 2024). Koopman autoencoders learn an encoder which maps a state to a latent representation, a matrix that advances that representation linearly, and a decoder that maps the latent state back to the original state space.

A central challenge of Koopman approximations is that the most interesting nonlinear systems typically contain multiple equilibria or multiple basins of attraction, and theory shows that these systems cannot admit a single finite-dimensional global linearization (Lan and Mezić, 2013; Brunton et al., 2016; Pan and Duraisamy, 2024). For example, a damped mechanical system can settle into different wells; a biological or chemical model can approach different stable equilibria; a controlled system can move between regions with distinct local dynamics. In such multistable systems, it is possible to linearize the dynamics within basins, but the linearizations needed in different basins of attraction are incompatible with each other. Given the theoretical argument, multibasin forecasting with a single global finite-dimensional linearization seems ill-posed; it is more naturally a problem of discovering which local linear description, or which description valid across a basin, applies at the current state.

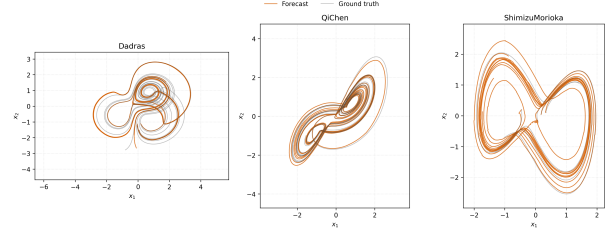
If Koopman models are to serve as a foundation for interpretable multi-regime modeling, then the latent representation of a Koopman autoencoder should do more than forecast well: ideally, it should organize the state space into meaningful local dynamical regimes. Our hypothesis is that sparse coding offers an appealing inductive bias for this purpose. Classical sparse coding and LISTA-style encoders produce piecewise-linear mappings and union-of-subspaces structure in latent space (Olshausen and Field, 1996; Chen et al., 1998; Donoho and Elad, 2003; Gregor and LeCun, 2010). If the latent representation of state produced by a LISTA encoder is sparse, then each state activates only a small subset of latent coordinates. That active subset, which we call a support, is a discrete object that can be inspected and used for routing to correct locally linear dynamics. Then, periodically decoding and re-encoding the state can refresh the quality of the latent embedding over time by re-selecting the correct local dynamics as trajectories move across the state space (Fathi et al., 2024).

This paper tests that hypothesis in a deliberately label-free learning setting. During training and deployment, the model is not told the number of basins, the basin assignment of any trajectory, or which local linear law should be used. Basin labels, when available in controlled benchmarks, are reserved for evaluation. The contribution is an empirical and mechanistic study of whether induced latent sparsity produces support objects that agree with basin labels in evaluation, whether those support objects can route local predictors without oracle labels, and whether the resulting sparse-latent Koopman models remain competitive on long-horizon forecasting tasks.

The paper follows this chain directly. [Section 2](#) formulates multibasin Koopman learning, [section 3](#) explains why sparse supports can identify local dynamical regimes, and [section 4](#) defines the sparse-latent model, support objects, re-encoding rule, and training objective. [Section 5](#) then states the evaluation protocol, including the support-routed prediction diagnostics, and [section 6](#) presents the evidence in the same order as the claim: agreement between supports and basin labels, non-oracle support-routed prediction, support refresh after transfer into a new basin, and long-horizon forecasting.



(a) Representative multibasin benchmark systems.



(b) Long-horizon Dysts forecasting examples at $H=20000$.

Figure 1: The paper evaluates two complementary regimes. The procedural multibasin systems provide known basin labels for interpretability diagnostics, while the Dysts systems provide long-horizon forecasting stress tests beyond the controlled low-dimensional setting.

2 Problem formulation

The introduction motivates a multibasin version of Koopman learning. Given sampled trajectories of a nonlinear system, the aim is to learn a state-reconstructing latent representation whose evolution is linear, without being told which basin or local dynamical regime generated each observation. The benchmarks may be defined by continuous-time dynamics, but the learner is only given stored observations. Thus, for each system, we work with the one-step observation map

$$x_{k+1} = F_{\Delta t}(x_k), \quad (1)$$

where $x_k \in \mathbb{R}^{d_x}$ is the state at the k th stored sample, $F_{\Delta t}$ denotes one benchmark observation step, and Δt is the system-specific observation interval. The model does not observe the continuous flow, the vector field, or the internal integration steps used to generate the cached trajectories. Forecasting is therefore defined at the stored sampling cadence: a horizon of H means H applications of $F_{\Delta t}$. Because Δt may differ across benchmark systems, equal values of H need not correspond to equal physical time.

Basins of attraction. Many systems of interest are multistable: different initial conditions can approach different long-run behaviours. A fixed point is a state x^* satisfying $F_{\Delta t}(x^*) = x^*$, equivalently $f(x^*) = 0$ in continuous time. More generally, a trajectory may approach an attractor A , such as a stable equilibrium, limit cycle, torus, or chaotic invariant set. The basin of attraction of A is the set of initial conditions whose trajectories converge to A :

$$\mathcal{B}(A) = \left\{ x_0 \in \mathbb{R}^{d_x} : \text{dist}\left(F_{\Delta t}^k(x_0), A\right) \rightarrow 0 \text{ as } k \rightarrow \infty \right\}. \quad (2)$$

When multiple attractors coexist, their basins partition the state space up to basin boundaries. This creates the regime of interest for this paper. Within one basin, a useful linear description may exist locally or even basin-wide in appropriate coordinates, but another basin may require an incompatible description. Thus, the task is to learn a representation that can support linear prediction while the appropriate linear law may depend on where the current state lies in the multibasin geometry.

Koopman autoencoders. We study this setting with Koopman autoencoders. The model learns an encoder $E_\theta : \mathbb{R}^{d_x} \rightarrow \mathbb{R}^{d_z}$, a decoder $D_\phi : \mathbb{R}^{d_z} \rightarrow \mathbb{R}^{d_x}$, and a latent linear transition matrix $K \in \mathbb{R}^{d_z \times d_z}$. Given an initial observation x_k , the corresponding open-loop h -step forecast is

$$\hat{x}_{k+h} = D_\phi \left(K^h E_\theta(x_k) \right). \quad (3)$$

The same matrix K is applied at every step of the rollout, so any state-dependent structure required for prediction must be encoded in the learned representation rather than in a supplied regime label.

Training windows. Training uses windows of adjacent stored observations. For a batch of B windows of rollout length L , $\{x_{b,k}, x_{b,k+1}, \dots, x_{b,k+L}\}_{b=1}^B$, where b indexes windows and ℓ indexes offsets within a window, the model encodes each observed state, rolls out from the initial encoded state, and decodes both reconstructed observations and latent forecasts. The encoded/reconstructed quantities are defined for $\ell = 0, \dots, L$, while rollout/forecast quantities are defined for $\ell = 1, \dots, L$:

$$\begin{aligned} z_{b,k+\ell}^{\text{enc}} &= E_\theta(x_{b,k+\ell}), & z_{b,k+\ell}^{\text{roll}} &= K^\ell z_{b,k}^{\text{enc}}, \\ x_{b,k+\ell}^{\text{rec}} &= D_\phi(z_{b,k+\ell}^{\text{enc}}), & \hat{x}_{b,k+\ell} &= D_\phi(z_{b,k+\ell}^{\text{roll}}). \end{aligned} \quad (4)$$

The training losses described later are built from these reconstructed and rolled-out quantities. Each application of K is interpreted as one stored benchmark step, matching the observation map in [equation \(1\)](#).

Model discovers basins unsupervised. In the intended training and deployment setting, the model is not given basin labels, the number of basins, or trajectory-to-basin assignments. When benchmark basin annotations are available, they are reserved for post-hoc evaluation; they are not used to choose the model, train the encoder or transition, or route predictions.

3 Sparse supports can identify local regimes

[NOTE: Maybe throw in some math and/or algorithms here to make this look more sophisticated. There is a good story-telling here but maybe this also overlaps with Introduction somewhat. Also, need to increase information density.]

If the same latent transition matrix is applied throughout state space, then the representation itself must carry whatever information is needed to distinguish one local dynamical law from another. A generic dense encoder can, in principle, store this information in its continuous coordinates, but it has no particular reason to make the distinction inspectable or reusable for routing. For multibasin systems, we would like a representation with both continuous coordinates for prediction and a discrete, data-driven indicator of which local dynamical regime is being used. Sparse coding provides exactly this kind of inductive bias: the values of the active coordinates describe the state within a local regime, while the active set identifies that regime.

The basic sparse coding model makes this geometry explicit. Given a dictionary decoder $D_\phi : \mathbb{R}^{d_z} \rightarrow \mathbb{R}^{d_x}$, sparse coding seeks a representation z for input state x such that $x \approx D_\phi z$ by solving

$$\min_z \frac{1}{2} \|x - D_\phi z\|_2^2 + \lambda \|z\|_1, \quad (5)$$

where $\lambda > 0$ controls how many coordinates are active ([Olshausen and Field, 1996](#); [Chen et al., 1998](#); [Donoho and Elad, 2003](#)). The important point is that the support z is a sparse, structured object. As x varies within a region where the same coordinates remain active, the sparse code

varies smoothly, often affinely, within the span of a fixed set of dictionary atoms. When x crosses into a region better explained by different atoms, the support can change. Sparse coding therefore naturally represents data as a union of lower-dimensional linear pieces indexed by active sets.

[TODO: Consider adding a figure here to illustrate the point, or illustrate the overall theme of the section.]

This geometry matches the multibasin problem well. A basin, or a coherent part of a basin, is precisely a region where one expects a local predictive description to be more stable than a global one. If the encoder assigns nearby states in that region to the same or closely related supports, then the support can act as a learned chart indicator. If the encoder assigns nearby states in that region to the same or closely related supports, then the support gives a learned, inspectable cue for the relevant basin or local predictive description. The continuous latent coefficients can still carry within-region state information, while the support can be inspected, compared across trajectories, or used to select a local predictor.

[NOTE: might have to compress or remove this paragraph] There are two distinct claims here. The first is static: observing the support should reduce uncertainty about which basin or coherent region contains the state. The second is dynamical: the same support should identify which latent coordinates, or which fitted local linear law, should be used for prediction. The first claim does not imply the second. An encoder could use some active coordinates as markers of basin membership while the one-step dynamics are carried by other coordinates, by the continuous coefficient values, or by cross-coordinate couplings in the learned transition matrix. The experiments therefore test support-label agreement and support-routed prediction separately. The overarching hypothesis is that sparsity can produce support families that mostly avoid mixing basin labels and are useful enough to route local prediction without oracle labels.

LISTA as a fast sparse encoder. LISTA is a practical way to impose this bias inside a neural Koopman model. Classical sparse coding solves the above optimization with an iterative algorithm (e.g., ISTA/FISTA, coordinate descent) [TODO: citations for ISTA/FISTA/coordinate descent], which is awkward and slow. LISTA approximates the iterative solver’s sparse solution with a learned finite-depth shrinkage network (Gregor and LeCun, 2010). The thresholding operations make the map from state to latent code explicitly support-selecting, while the surrounding learned weights allow the sparse inference procedure to adapt to the data and forecasting objective. In this paper, LISTA is one structured implementation of induced sparsity; sparse MLP controls test whether the decisive ingredient is the sparse support itself rather than the particular encoder architecture.

[TODO: merge the more technical descriptions on LISTA into previous paragraph or just forget it. There should be at most one or two paragraphs on LISTA. We could use appendices. See commented out text]

4 Method

4.1 Operationalizing the support tests

[NOTE: We don’t need to do this now, but we might need to shorten this subsection significantly or remove it entirely.]

The preceding sections reduce the problem to a concrete question: can a model trained without basin supervision learn a discrete, inspectable object that identifies the relevant basin or local

dynamical regime? We separate this question into the following chain:

$$\text{sparse latent code} \implies \text{support agrees with basin labels} \implies \text{support selects local prediction.} \quad (6)$$

The model and objective induce the first term. The two arrows are then evaluated after training. The first arrow asks whether support objects agree with benchmark basin labels when those labels are used only as an evaluation overlay. The second arrow asks whether the same support objects can select useful local predictive laws without replacing them by oracle basin assignments. This distinction is important: the learning problem remains label-free, while the benchmark labels let us measure whether the discovered discrete structure corresponds to the known multibasin geometry. It also prevents a false shortcut in the interpretation: a support can be an excellent region label and still fail as a predictor selector if the columns of the learned Koopman transition that matter locally are not the columns marked active by that support.

4.2 Support objects

The central object in this paper is not the raw latent vector alone, but the active set it induces. Given an encoded state $z = E_\theta(x)$, a support rule converts z into a binary mask or fixed-size active set. We use three support rules:

- **Absolute-threshold support.** $S_{\text{abs}}(x) = \{i : |z_i| > 10^{-3}\}$. This asks the most literal identifiability question: deep inside a basin, does one exact active set dominate?
- **Relative-threshold support.** $S_{\text{rel}}(x) = \{i : |z_i| > 0.1 \max_j |z_j|\}$. This removes global scale effects and is useful when comparing local linear laws.
- **Fixed-size top-8 support.** $S_{\text{top8}}(x) = \text{top}_8(|z|)$, where $\text{top}_8(|z|)$ denotes the indices of the eight largest-magnitude latent coordinates. This is the most deployment-like object because it avoids choosing an activity threshold and assigns every state the same support size.

We use these rules to define two kinds of support objects:

- **Exact support.** The exact support is the set produced directly by one of the support rules above. It is the stringent object used to test whether a basin can be represented by a single active mask.
- **Support family.** A support family groups nearby exact supports by Jaccard overlap, using a threshold of 0.5 against the family prototype. This asks whether apparently different masks are small variations of the same support associated with one local dynamical regime, rather than treating every threshold crossing as a new regime. During family-routed prediction, an exact support observed at test time is assigned to a previously fitted family only if it matches an existing family closely enough; otherwise the route is treated as unavailable and the predictor falls back to the global law.

The distinction is necessary because the strongest plausible claim is not that every basin must correspond to one exact mask everywhere. A basin may be represented by several nearby masks, especially near separatrices, in regions where a coordinate is weakly active, or under equivalent latent coordinate choices. Exact support uniqueness is therefore a stringent diagnostic, while a support family is the more robust object for asking whether the representation agrees with the multibasin geometry. In block-structured variants we may also record the dominant latent group, but the target claim concerns active supports, not pre-specified latent blocks.

4.3 Sparse-latent Koopman model

The model that produces these support objects is a Koopman autoencoder with encoder E_θ , decoder D_ϕ , and latent transition K . In the linear-decoder variants, the basic maps are

$$z_t = E_\theta(x_t) \in \mathbb{R}^{d_z}, \quad D_\phi(z) = Dz, \quad \hat{z}_{t+1} = Kz_t, \quad \hat{x}_{t+1} = D_\phi(\hat{z}_{t+1}). \quad (7)$$

Thus an active coordinate can be read as selecting a learned dictionary direction, and between re-encoding events the latent state is advanced by the same linear transition. Sparsity is therefore a property of the learned representation, not an assumption that the physical state, the vector field, or the Koopman transition itself is sparse.

In experiments, Koopman transition structure is varied separately from the encoder. This distinguishes the effect of induced sparsity from the effect of the LISTA inference architecture or the transition structure alone.

4.3.1 LISTA sparse encoder

Our main sparse encoder uses LISTA-style amortized sparse inference. The encoder first computes a dense pre-code $c_t = f_\theta(x_t)$ and then applies learned shrinkage refinements, for $q = 0, \dots, Q - 1$:

$$u_t^{(0)} = \text{shrink}(c_t, \tau), \quad u_t^{(q+1)} = \text{shrink}(Su_t^{(q)} + c_t, \tau), \quad z_t = \Pi(u_t^{(Q)}). \quad (8)$$

Here $\text{shrink}(a, \tau) = \text{sign}(a) \max(|a| - \tau, 0)$ is the standard soft-thresholding shrinkage map applied elementwise, S is learned, τ is a learned or tuned threshold scale, and Π denotes the final coordinate convention, such as a nonnegative code or a sign-split code. [TODO: is this true? fact check this] Shrinkage gives the encoder an explicit mechanism for selecting supports, while the learned pre-code and refinement weights allow that selection rule to adapt to the dynamical prediction task.

4.3.2 Koopman transition structure

[TODO: can make this more concise, perhaps by using bullets. don't need thorough explanations, just one explanation for differences between all three.]

Dense transition: all latent coordinates may interact. The dense transition leaves K unrestricted:

$$K_{\text{dense}} \in \mathbb{R}^{d_z \times d_z}. \quad (9)$$

This is the least structured latent dynamics model. It can use any column of K to influence any latent coordinate, so support-label agreement does not by itself imply that active coordinates form an isolated local predictor.

Block-diagonal transition: no cross-group latent coupling. The block-diagonal transition partitions latent coordinates into M groups and constrains

$$K_{\text{block}} = \text{blkdiag}(K_1, \dots, K_M). \quad (10)$$

Here each group evolves internally, with no learned coupling from one group to another. The groups are architectural coordinates, not supplied basin assignments; the experiments still ask whether the learned supports agree with basin labels after training.

Soft-block transition: cross-group coupling is discouraged, not forbidden. The soft-block transition uses the same latent grouping but keeps K dense. Instead of zeroing cross-group entries, it adds a penalty

$$\mathcal{L}_{\text{offblock}} = \sum_{i,j: g(i) \neq g(j)} K_{ij}^2, \quad (11)$$

where $g(i)$ denotes the latent group containing coordinate i . This variant tests whether mild pressure toward group-local dynamics is enough, while still allowing the model to learn cross-group interactions when they improve prediction.

4.4 Refreshing the support during rollout

If a support is interpreted as an indicator of the currently relevant basin or local dynamics, then a long rollout should be allowed to re-select the support as the predicted state evolves. We therefore evaluate periodic re-encoding, following Fathi et al. (2024). Starting from an encoded state $z_k = E_\theta(x_k)$ and a re-encoding period m measured in stored observation steps, the model advances for m Koopman steps, decodes the predicted state, and then encodes that prediction again:

$$\tilde{z}_{k+m} = K^m z_k, \quad \tilde{x}_{k+m} = D_\phi(\tilde{z}_{k+m}), \quad z_{k+m} = E_\theta(\tilde{x}_{k+m}). \quad (12)$$

The same procedure is repeated over the rollout horizon. In the forecasting summaries we evaluate a discrete set of candidate periods and report the best-periodic score over that set, but each individual rollout uses only the model’s own predicted state. It does not use the true future state, a basin label, or an oracle for transitions between basins. Its main role is to let the encoder re-evaluate which support should be active at the current predicted location.

[TODO: This paragraph on periodic grid should be moved to experimental procedure or removed.] The default periodic grid is system-dependent. For the fixed multibasin benchmark the standard candidate periods are 10, 25, 50, and 100 stored observation steps, with every-step re-encoding reported separately. For the Dysts long-horizon benchmark we use the extended grid 1, 5, 10, 20, 40, 60, 80, 100, 200, 300, 400, 500, 1000 because the evaluated horizons are much longer. The best-periodic number is therefore a model-selection diagnostic over a fixed evaluation grid, not a rollout that adapts its reset period online.

4.5 Training objective

Training uses only adjacent observation windows. For a batch of B windows with prediction horizon L (and therefore $L + 1$ observed states), the model produces reconstructed future states $x_{b,k+\ell}^{\text{rec}}$, latent rollouts $z_{b,k+\ell}^{\text{roll}}$, encoded future latents $z_{b,k+\ell}^{\text{enc}}$, and decoded rollout predictions $\hat{x}_{b,k+\ell}$ as defined in equation (4). The rollout is seeded by the initial state, and the reconstruction, prediction, latent-linearity, and sparsity sums below are over the future offsets $\ell = 1, \dots, L$. The main losses are

$$\bar{\mathcal{L}}_{\text{pred}} = \frac{1}{BL\sqrt{d_x}} \sum_{b,\ell} \|\hat{x}_{b,k+\ell} - x_{b,k+\ell}\|_2, \quad \bar{\mathcal{L}}_{\text{rec}} = \frac{1}{BL\sqrt{d_x}} \sum_{b,\ell} \|x_{b,k+\ell}^{\text{rec}} - x_{b,k+\ell}\|_2, \quad (13)$$

$$\bar{\mathcal{L}}_{\text{lin}} = \frac{1}{BL} \sum_{b,\ell} \|z_{b,k+\ell}^{\text{roll}} - z_{b,k+\ell}^{\text{enc}}\|_2, \quad \bar{\mathcal{L}}_{\text{sp}} = \frac{1}{BL} \sum_{b,\ell} \|z_{b,k+\ell}^{\text{roll}}\|_1. \quad (14)$$

The observation-space terms are normalized by $\sqrt{d_x}$ so that their scale is comparable across systems of different state dimension. The latent terms are left in their native scale because the latent dimension is fixed within each comparison.

The total objective is

$$\mathcal{L} = \frac{1}{L} (\lambda_{\text{pred}} \bar{\mathcal{L}}_{\text{pred}} + \lambda_{\text{rec}} \bar{\mathcal{L}}_{\text{rec}} + \lambda_{\text{lin}} \bar{\mathcal{L}}_{\text{lin}} + \lambda_{\text{sp}} \bar{\mathcal{L}}_{\text{sp}}) + \mathcal{L}_{\text{struct}}. \quad (15)$$

Here $\mathcal{L}_{\text{struct}} = 0$ for the dense and block-diagonal transition variants. For the soft-block transition, $\mathcal{L}_{\text{struct}} = \lambda_{\text{off}} \mathcal{L}_{\text{offblock}}$, with $\mathcal{L}_{\text{offblock}}$ defined above as the penalty on cross-group entries of K . The additional horizon normalization keeps longer training windows from automatically increasing the scale of the sequence terms. The prediction and latent-linearity terms make the representation useful for Koopman rollout, the reconstruction term keeps the code tied to the observed state, and the sparsity term encourages selective activation. No loss term uses basin labels, basin counts, or trajectory-to-basin assignments.

5 Experimental procedure

The experiments instantiate the chain in [equation \(6\)](#): sparse latent code $\Rightarrow$ support agrees with basin labels $\Rightarrow$ support selects local prediction. We evaluate the first arrow (alignment) and the second arrow (routing) on a controlled multibasin benchmark whose endpoint basin labels are revealed only at evaluation, and re-test the forecasting half of the chain on a chaotic long-horizon benchmark where basin labels do not exist.

5.1 Benchmarks

Multibasin benchmark. A fixed suite of 17 procedurally generated two-dimensional systems exposes diverse basin geometries: triple-well and quadratic-coupling potentials, gated local-linear and gated-transfer-linear systems, polygonal multi-attractor arrangements with 3–10 basins, checkerboard potentials, depth-cascade and asymmetric variants, and transition-rich layouts in which finite-horizon trajectories pass close to multiple basins. Endpoint basin labels exist for evaluation but never enter training, model selection, support extraction, or routing. [Figure 1\(a\)](#) shows representative members; the full inventory is in [section B](#).

Long-horizon chaotic benchmark. 15 chaotic systems from the Dysts repository ([Gilpin, 2021, 2023](#)) stress-test forecasting at horizons $H5000$ – $H30000$, well beyond the regime in which a single linear latent transition stays accurate without re-encoding. Basin labels are unavailable, so this benchmark probes forecasting only.

5.2 Models

We compare five trained roots throughout the multibasin benchmark, all under the same boundary-emphasized sampling regime that oversamples initial conditions near separatrices to expose ambiguous-basin states:

- **LISTA-BD**: LISTA encoder with sign-split code, block-diagonal K ($M=8$ blocks of size 4).
- **LISTA-SB**: LISTA encoder with sign-split code, dense K with the soft-block penalty $\mathcal{L}_{\text{offblock}}$ at $\lambda_{\text{off}}=10^{-4}$ and $d_z=64$.
- **Sparse MLP**: dense MLP encoder with ℓ_1 penalty on z , dense K .
- **Sparse MLP, BD**: dense MLP encoder with ℓ_1 penalty, block-diagonal K .

- **Dense MLP:** dense MLP encoder with no ℓ_1 penalty (no shrinkage). The strongest non-sparse comparator and the falsification anchor.

The two LISTA roots and the two sparse-MLP roots share ℓ_1 shrinkage and sampling, so their head-to-head comparison isolates the encoder architecture; the Dense MLP isolates the contribution of induced sparsity itself. Latent dimensions are matched within each comparison ($d_z=64$ for LISTA-SB, $d_z=256$ otherwise). Training uses [equation \(15\)](#) with $\lambda_{\text{pred}}=0.03$, $\lambda_{\text{rec}}=1$, $\lambda_{\text{lin}}=0.03$, $\lambda_{\text{sp}} \in \{0, 3 \times 10^{-3}, 6 \times 10^{-3}\}$, sequence length $L=8$, and 200,000 Adam steps with a cosine schedule. Each (root, system) pair uses 10 random seeds. The Dysts experiments use the same recipe with standard initial conditions and add a dense-transition LISTA root (**LISTA-D**) and two sparsity strengths of LISTA-BD as fairness controls.

5.3 Forecasting metric and statistical protocol

For each trained checkpoint we generate held-out trajectories and evaluate two rollout schemes. *No re-encode* applies K^h once and decodes; *periodic* decodes/re-encodes every m stored steps over a discrete grid ([equation \(12\)](#)): $\{10, 25, 50, 100\}$ on the multibasin benchmark and $\{1, 5, 10, 20, 40, 60, 80, 100, 200, 300, 400, 500, 1000\}$ on Dysts. The headline error is the minimum over this grid (best-periodic), reported alongside the no-reencode column in the appendix.

The headline test is per-system rather than aggregated. Within each of the 17 benchmark systems we have 10 paired (candidate, baseline) seeds; we run a one-sided paired Wilcoxon signed-rank test on $\log_{10}$ -MSE differences against the matched-sampling Dense MLP, Holm-corrected across the 17 systems at $\alpha=0.05$. Wilcoxon is preferred because catastrophic seeds make the within-system paired-difference distribution non-Gaussian; a paired t -test sanity check agrees with Wilcoxon on ≥ 14 of 17 systems in every cell. The reported headline is therefore K/N : the number of systems on which the candidate beats the baseline after Holm correction. Per-cell point estimates are interquartile means (IQM) over the 10 seeds, with a cross-system IQM as the headline value and a $\log_{10}$ -IQR over the 17 system points as the dispersion summary. The same per-system framework is applied to the interpretability columns ($|S|$, $H(B | F)$, wrong-support-freeze ratios); $H(B | S)$ saturates to 0 on the deep slice for every row and is therefore not testable.

5.4 Support objects, interpretability, and routing

We extract three support objects from a trained encoder, evaluated on held-out trajectories. *Absolute-threshold* (S_{abs} , $\tau=10^{-3}$) is the strict identifiability object used on the deep-state slice (top quartile by basin-depth margin). *Relative-threshold* (S_{rel} , $r=0.1$) absorbs scale. *Top-8* (S_{top8}) is the deployment-like object because every state receives a route key without a magnitude threshold. Exact supports are merged into *support families* by greedy Jaccard merging at threshold 0.5.

Mean active size $|S| = \mathbb{E}_x[\#\{i : |z_i(x)| > \tau\}]$ measures support compression on the deep slice; $|S| \ll d_z$ is necessary for any basin-identifiability claim to be non-vacuous. Basin uncertainty given the exact support $H(B | S)$ and given the family $H(B | F)$ measure how sharply the support determines the basin. The wrong-support-freeze ratio ([equation \(16\)](#) with a canonical mask drawn from a different basin) measures whether the predictor actually *uses* the support: wrong-frozen / base MSE at rollout horizon h is large when the predictor relies on the right support and close to 1 when it does not. The diagnostic requires at least two distinct deep-slice basins, so it is undefined on 4 of the 17 systems whose deep slice is single-basin (denominator $K/13$).

For the routing test we freeze the trained encoder/decoder/ K , fit centered local laws on a disjoint operator-fit trajectory set, and roll out from the initial observation alone; the route at every step is the model’s own predicted top-8 support, so basin labels are not used. We compare three local

predictors against the global K baseline (equation (3)). The first masks the centered input by the current support:

$$\hat{z}_{t+1} = \bar{z}_c + K(m_c \odot (z_t - \bar{z}_c)), \quad (16)$$

where c is the support object inferred from the current latent state, m_c its binary mask, and $\bar{z}_c$ the corresponding latent center. The second fits ridge-regularized centered local slopes for each support object c :

$$A_c = \arg \min_A \sum_{t \in \mathcal{I}_c} \|(z_{t+1} - \bar{z}_c) - A(z_t - \bar{z}_c)\|_2^2 + \eta \|A\|_F^2, \quad (17)$$

predicting $\bar{z}_c + A_c(z_t - \bar{z}_c)$. The third is the family-level analog with the support replaced by its family. A route is considered available only when the inferred support matches a fitted class with ≥ 32 transitions; otherwise the predictor falls back to the global K . We report the median routed/global $H1000$ MSE ratio (lower is better; < 1 means routing helped) and the win rate (fraction of rollouts with ratio < 1); a one-sided binomial test on the win rate against $p_0=0.5$ provides per-cell significance.

Periodic support refresh. The refresh test uses controlled state-space transfers between known basins. A fitted forward simulator brings each trajectory from a source basin to inside a target basin (basin labels are used only to score post-entry behaviour). After entry we compare a *frozen-source* rollout (keep the support inferred at the source) to a *refreshed-current* rollout (re-encode every $m \in \{1, 10\}$ stored steps and re-extract the support). We report target-support dominance, route-target fraction, and the ratio of forecast MSE between refreshed-current and frozen-source rollouts.

Falsification controls. Three controls accompany every routing claim: (i) oracle-conditioned partitions (basin label supplied), (ii) random count-matched partitions, and (iii) latent k -means clusters with the same number of classes. We additionally compare each fitted A_c against the true Jacobian along sampled states (section F).

6 Results

We follow the chain in order. Section 6.1 shows that sparse-latent models produce small support objects whose mask determines the basin on states far from boundaries. Section 6.2 shows the support is functionally used by the predictor. Section 6.3 shows that this representation is also an excellent forecaster, and that induced sparsity rather than the LISTA encoder is the active ingredient. Section 6.4 and section 6.5 establish the deployment-time routing and refresh mechanism. Section 6.6 extends the forecasting story to long horizons on chaotic flows. Section 6.7 states the calibrated reading from the falsification diagnostics.

6.1 Sparse supports compress basin labels on states far from boundaries

Table 1 reports support-label agreement on the deep-state slice (S_{abs}) for all five trained roots. Basin-support alignment is the conjunction of two requirements: an active support is unambiguous about which basin produced it ($H(B | S)=0$), and the active support is small ($|S| \ll d_z$). Both halves are necessary: a "support" containing every latent coordinate trivially determines the basin because it contains the entire encoded state; this is not an interpretability claim.

Every sparse row reaches $H(B | S)=0$ on the deep slice, and the Dense MLP reaches $H(B | S) \approx 0.001$; numerically these are equivalent. The actual separation appears in the support-size

column. The Dense MLP achieves its near-zero $H(B | S)$ with $|S|=254$ active coordinates out of 256, i.e. essentially the full latent vector, which renders its alignment claim vacuous. LISTA-SB reaches $H(B | S)=0$ with only $|S|=27$ out of 64 active coordinates and produces only ~ 8 unique exact supports across all evaluated states – the strongest non-vacuous form of the basin-identifiability claim in the table. The two sparse-MLP rows compress to $|S|\sim 106$ out of 256, less than half the no-shrink baseline. Per-system Wilcoxon on $|S|$ rejects on 17/17 systems for every sparse row.

A more deployment-realistic version of the same claim asks whether the basin is determined by the *support family*. Every sparse row reaches $H(B | F) \leq 0.06$ with ~ 3 family clusters and clears Holm correction on 11–13 of 17 systems against the baseline. The Dense MLP, by contrast, collapses to a single family across all evaluated states ($H(B | F)=0.90$): its supports do not cluster into basin-distinguishing families at all. This is the cleanest cross-architecture interpretability gap in the table.

Table 1: Multibasin forecasting and support–basin agreement on the 17-system benchmark; all rows trained under matched boundary-emphasized sampling. Each cell shows the metric followed by $[K/N]$, the number of systems on which the row beats the Dense MLP baseline at Holm-corrected $\alpha=0.05$ (per-system one-sided paired Wilcoxon over 10 seeds; [section 5.3](#)). Forecasting columns are cross-system IQM of per-system IQM-over-seeds in raw MSE. The wrong-support-freeze columns use a $K/13$ denominator because the diagnostic is undefined on 4 systems whose deep slice is single-basin. Underlined entries are best in column.

Model	H100↓	H500↓	H1000↓	$ S $ ↓	$H(B S)$ ↓	$H(B F)$ ↓	wr. $h=1$ ↑	wr. $h=20$ ↑
LISTA-BD	0.0175 [14/17]	0.0682 [14/17]	0.0699 [15/17]	101 [17/17]	0	0 [13/17]	8.1×10^3 [13/13]	16.4 [10/13]
LISTA-SB	0.0255 [15/17]	0.0607 [14/17]	0.0708 [15/17]	27 [17/17]	0	0.055 [11/17]	3.0×10^3 [13/13]	8.22 [7/13]
Sparse MLP, BD	<u>0.0163</u> [15/17]	0.0513 [15/17]	<u>0.0452</u> [17/17]	106 [17/17]	0	0.00043 [12/17]	6.3×10^3 [13/13]	35.2 [12/13]
Sparse MLP	0.0183 [14/17]	<u>0.0421</u> [15/17]	0.0465 [17/17]	106 [17/17]	0	0 [13/17]	6.5×10^3 [13/13]	35.3 [12/13]
Dense MLP, no shrink	0.665 [baseline]	2.68 [baseline]	3.07 [baseline]	254 [baseline]	0.0012	0.9 [baseline]	8.68 [baseline]	1.35 [baseline]

6.2 The support is functionally used by the predictor

A separate question, distinct from “does the support identify the basin,” is “is the support actually used by the predictor.” [Table 1](#) answers it with the wrong-support-freeze diagnostic: take the model’s own predicted support, replace it with a canonical mask drawn from a *different* basin, and freeze that wrong mask for h rollout steps. The reported ratio is wrong-frozen MSE / unfrozen base MSE at the decoded prediction step.

For the four sparse rows the ratio is $3,000\text{--}8,100\times$ at $h=1$ and $8.2\text{--}35.3\times$ at $h=20$: catastrophic degradation when forced through the wrong support. For the Dense MLP it is $8.7\times$ at $h=1$ and $1.35\times$ at $h=20$: the baseline barely notices the swap. The per-system Wilcoxon clears Holm correction on 13/13 testable systems at $h=1$ and on 7–12 of 13 at $h=20$. The sparse rows therefore not only *produce* a basin-identifying small-support code, they *route through* it – the property the deployment-time routing experiments build on.

[Figure 2](#) shows the per-(system, seed) distributions of the standard exact-support metrics; appendix tables [tables 9](#) to [12](#) provide the 17-system $\times$ 4-candidate breakdown.

6.3 Sparse models forecast better than the dense baseline at every horizon

Interpretability matters only if the model is also a competent forecaster. [Table 1](#) shows the four sparse rows at $H1000$ raw MSE in $0.045\text{--}0.071$, while the matched no-shrink Dense MLP collapses to $H1000=3.07$. [Figure 3](#) traces the same gap across horizons and [figure 4](#) shows that the Dense MLP carries a much heavier upper tail at every horizon.

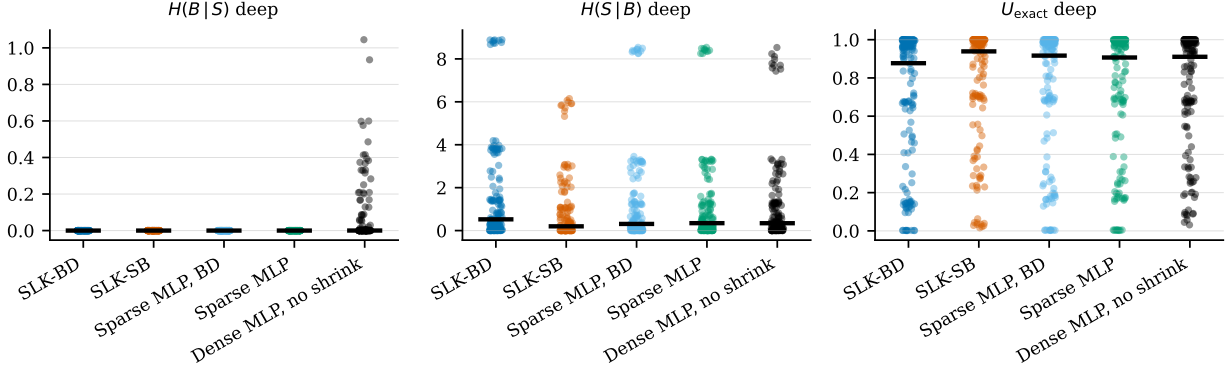


Figure 2: Per-(system, seed) basin/support entropy and dominant-support mass on the deep-state slice (S_{abs}). Each dot is one (system, seed) pair; black bar is the cross-system IQM. Lower is better for $H(B|S)$ and $H(S|B)$; higher is better for U_{exact} .

The per-system test sharpens this. At $H=1000$ (figure 5, table 8), the two sparse-MLP controls beat the no-shrink baseline on *all* 17/17 systems (per-system median effect -2.10 and -2.02 in $\log_{10}$ -MSE units, i.e. roughly $100\times$ lower MSE); LISTA-BD on 15/17 (median -1.86); LISTA-SB on 15/17 (median -1.36). The two systems where LISTA-BD and LISTA-SB miss Holm correction are `checkerboard_potential` (LISTA-BD: a wide-CI seed kept the rank statistic below threshold despite a large median effect) and `multiwell_strong_transition` / `arrested_spiral` (LISTA-SB: catastrophic seeds drove rank variability). At $H100$ and $H500$ the same per-system testing yields $K/N \in [14, 15]/17$ for every sparse row.

Induced sparsity, not the LISTA encoder, is the active ingredient. The Sparse MLP and Sparse MLP, BD controls reach essentially the same forecasting curve as LISTA-BD and LISTA-SB and produce the same qualitative interpretability metrics in table 1. This isolates the ℓ_1 penalty itself: induced latent sparsity, not the LISTA encoder, is the inductive bias that drives long-horizon stability under the forecasting objective. The LISTA encoder has a smaller, second-order effect, principally on support fragmentation: LISTA-SB has the lowest fragmentation IQM and is the only row whose $H(S|B)$ improvement over the no-shrink baseline survives per-system Wilcoxon (on 6/17 systems). The forest plot also shows that for any single sparse-vs-sparse pair (e.g. LISTA-BD vs. Sparse MLP, BD), most systems have overlapping 95% bootstrap CIs at $H1000$, consistent with the lack of an architecture-only effect once sampling is matched.

6.4 Supports route useful local predictors without oracle labels

If the inferred support identifies the basin and is used by the predictor, then it ought to stand in for the basin label and select a useful local law at deployment time. Table 2 reports this test. We freeze the trained encoder/decoder, fit centered local laws on a disjoint operator-fit trajectory set, and roll out from the initial observation alone; the route at every step is the model’s own predicted top-8 support, so no basin labels are used.

We read significance within each (model, slice, route) cell as a one-sided binomial sign test on the per-system IQM of $\log(\text{routed}/\text{global})$ ratios: how many of the ≤ 17 testable systems have IQM-delta below zero (favouring routing). This is the appropriate within-system reading because per-system seed counts in the routing CSV are too small (1–10 valid pairs per system) to clear Holm-corrected $\alpha/17 \approx 0.003$ on a per-system paired Wilcoxon: at the Table-1 standard, paired

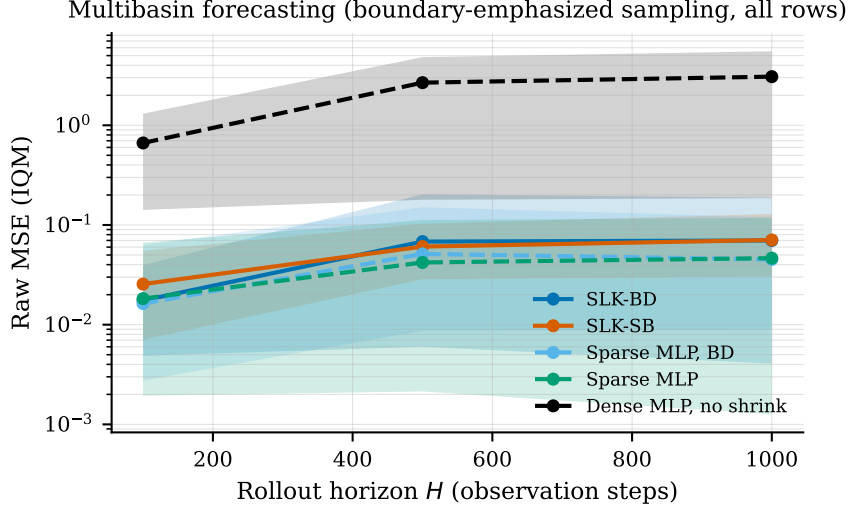


Figure 3: Multibasin forecasting horizon curves. Lines are cross-system IQM of per-system IQM-over-seeds; bands are the 25–75 system percentile range. The Dense MLP control degrades by more than an order of magnitude at every horizon, while the four sparse rows are statistically indistinguishable in IQM and only differ at the per-system level (figure 5).

Wilcoxon clears Holm on at most 0–3 of 17 systems per cell regardless of effect, so the sign test on per-system IQM-deltas is the more sensitive within-system stat at this design.

LISTA-SB beats its global K on every testable system for support-gated and centered support-local routing (sign-test 14/15 and 15/15, $p \leq 5 \times 10^{-4}$), with cross-rollout IQM ratios of 0.25 and 0.30 on the full slice. The cell IQMs sharpen on the deep-state slice (0.30 and 0.36) with sign-test 15/15 in both. The family-local route trims the IQM further to 0.11/0.12 at the cost of a minority of catastrophic rollouts (long right tail in figure 6); the within-system sign test still rejects at 13/15 ($p = 3.7 \times 10^{-3}$). LISTA-BD shows a smaller, less directional support-gated/support-local margin (cell IQMs 0.7–0.85, sign-test 5–6 of 8, no rejection), but its family-local route is decisive: cell IQM near zero (~ 0.003 – 0.004) with 8/8 systems negative ($p = 3.9 \times 10^{-3}$). The Dense MLP control sits close to the no-improvement line for the gated and local routes (IQM 0.88–0.92, marginal sign-test 12–13/16, $p \in [0.01, 0.04]$), and its family-local route fails outright (IQM > 290 on both slices, 2/16 systems negative, $p = 1$ in the wrong direction) – the dense baseline’s supports do not cluster into coherent families.

The fixed-size top-8 support is the right deployment object: it produces a route key for every state without a brittle magnitude threshold, and it is the only support definition for which the Dense MLP control’s failure cannot be attributed to a route-availability artifact.

6.5 Refreshing the support tracks basin transfer

Routing shows that an inferred support carries enough information to select a useful predictor; the refresh test asks whether the rollout mechanism keeps that support current as the trajectory moves through state space. We construct controlled state-space transfers between known basins and compare a frozen-source rollout to a refreshed-current rollout under the same periodic re-encoding period (table 3).

LISTA-SB refreshed-current rollouts reach target-support dominance 0.83/0.86 at re-encode periods 1/10 and reduce the forecast error to under 1% of the frozen-source baseline (cross-rollout

Fixed-17 multibasin forecasting per (system, seed), boundary-emphasized rows

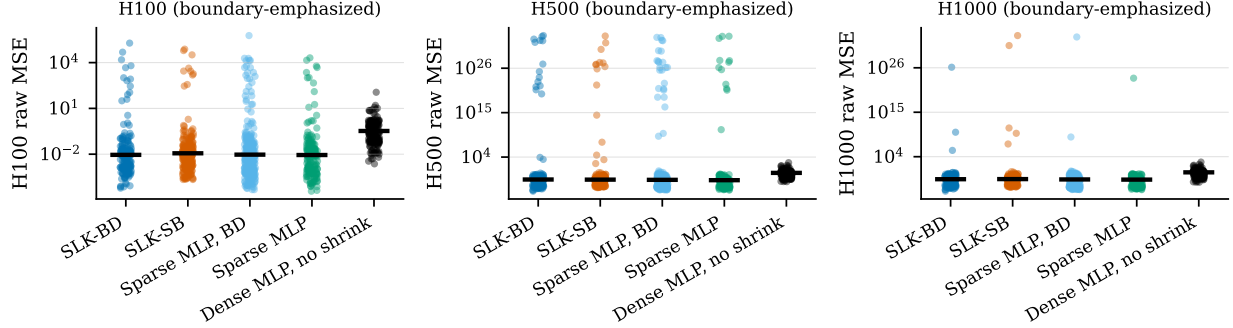


Figure 4: Per-(system, seed) forecasting MSE at $H100$, $H500$, and $H1000$ on the multibasin benchmark. Black bars are the IQM. The Dense MLP has a much heavier upper tail at every horizon.

Table 2: Non-oracle support-routed forecasting at $H1000$ on the multibasin benchmark, top-8 supports. The route at every step is the model’s own predicted top-8 support (no basin labels). Each cell is the cross-rollout IQM of the routed/global $H1000$ MSE ratio (lower is better; <1 means routing helped); rollout count per cell ranges from 23 to 158. Markers come from a one-sided binomial sign test on per-system IQM-deltas (does the per-system IQM of $\log(\text{routed/global})$ lie below zero?), evaluated over the 8–16 testable systems per (model, route) cell: $^\dagger p < 0.05$, $^\ddagger p < 10^{-3}$, no marker means $p \geq 0.05$. Bold marks the best IQM in each column.

Model	Slice	Support-gated K	Centered support-local	Centered family-local
LISTA-SB	all	0.25[‡]	0.30[‡]	0.11[†]
LISTA-SB	deep	0.30 [‡]	0.36 [‡]	0.12 [†]
LISTA-BD	all	0.78	0.73	0.004 [†]
LISTA-BD	deep	0.84	0.85	0.003[†]
Dense MLP	all	0.88 [†]	0.91 [†]	> 340
Dense MLP	deep	0.90 [†]	0.92 [†]	> 290

IQM of refreshed/frozen MSE: 0.0085 at period 1, 0.0074 at period 10). LISTA-BD has weaker exact-support dominance (0.07) because its block-diagonal supports change less when the latent moves; its route-target fraction is nevertheless 0.74/0.82, indicating that routing through the family is the correct grain to read LISTA-BD on this diagnostic. The cross-rollout IQM ratio is 0.58 (period 1) and 0.52 (period 10). The MSE-ratio gain is per-system robust: a one-sided paired Wilcoxon on $\log_{10}(\text{refreshed/frozen})$ within each system, Holm-corrected across the 12 systems with at least one transfer pair, rejects on 11/12 systems for LISTA-SB at both periods and on 9/12 (period 1) and 11/12 (period 10) systems for LISTA-BD. The companion sign test on per-system IQM-deltas adds direction: 11/11 testable systems negative for both LISTA-SB cells ($p=4.9 \times 10^{-4}$); 10/12 negative for LISTA-BD at period 1 ($p=1.9 \times 10^{-2}$) and 12/12 at period 10 ($p=2.4 \times 10^{-4}$). Figure 7 shows that both models keep refresh quality stable as the period varies between 1 and 10. The mechanism is therefore concrete: re-encoding lets the encoder re-extract a support that matches the local basin geometry, and the routed local law applied through that support is a much better predictor than the stale source support.

Per-system paired Wilcoxon effect sizes (10 seeds/system, Holm-corrected at $\alpha = 0.05$ across 17 systems)

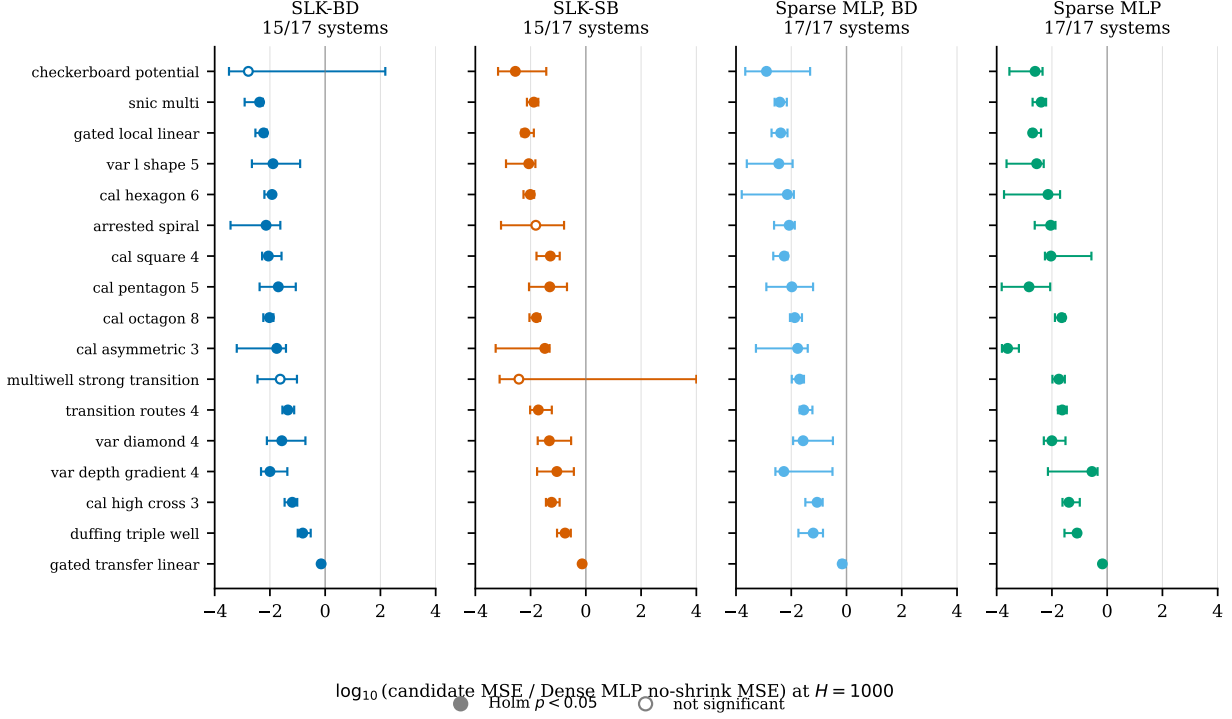


Figure 5: Per-system effect sizes at $H=1000$ versus the matched-sampling Dense MLP baseline. Each point is one of the 17 benchmark systems; x -coordinate is the per-system median paired $\log_{10}$ -MSE difference (candidate – baseline) over the 10 seeds, with whiskers the 95% bootstrap CI. Filled markers clear Holm-corrected $\alpha=0.05$. Title counts give the number of systems passing Holm correction. Numerical version: [table 8](#).

6.6 Long-horizon forecasting on chaotic flows

The multibasin benchmark is low-dimensional by design. To test whether sparse-latent models stay competitive outside it, we evaluate forecasting on 15 chaotic Dysts systems at horizons up to $H30000$ ([table 4](#), [figure 8](#)). The dense-transition LISTA root LISTA-D is strongest at the shorter horizons ($H5000/H10000$, IQM 0.166/0.957, lowest IQM on 12/11 of 15 systems). At $H30000$ the structurally constrained block-diagonal LISTA with stronger sparsity (LISTA-BD high sp) takes over, with $H30000$ IQM 2.235 and the lowest IQM on 9 of 15 systems.

The Dysts cross-system effects are intrinsically smaller than the multibasin gaps (per-horizon $\log_{10}$ -MSE differences of 0.05–0.2 rather than 1–2), so the per-system paired Wilcoxon vs the Dense MLP baseline – needing raw $p < 0.05/15 \approx 0.003$ to clear Holm correction – rejects on at most 1 of 15 systems per cell at the current $n_{\text{seed}}=10$ design (paired Wilcoxon’s exact p -floor at $n=5-10$ is $1/2^n$, which is at or above the Holm threshold for most cells). The within-system reading is therefore a one-sided binomial sign test on the per-system IQM of $\log_{10}$ -MSE deltas: LISTA-D wins on 12–13 of 15 systems at every horizon ($p \leq 0.018$); LISTA-BD (high sp) wins on 14/15 at $H20000$ and 14/15 at $H30000$ ($p \leq 5 \times 10^{-4}$); LISTA-BD (low sp) wins on 15/15 at $H30000$ ($p=3 \times 10^{-5}$) but *loses* at the shorter horizons (1–2/15 negative, $p < 0.005$ in the wrong direction). The MLP variants

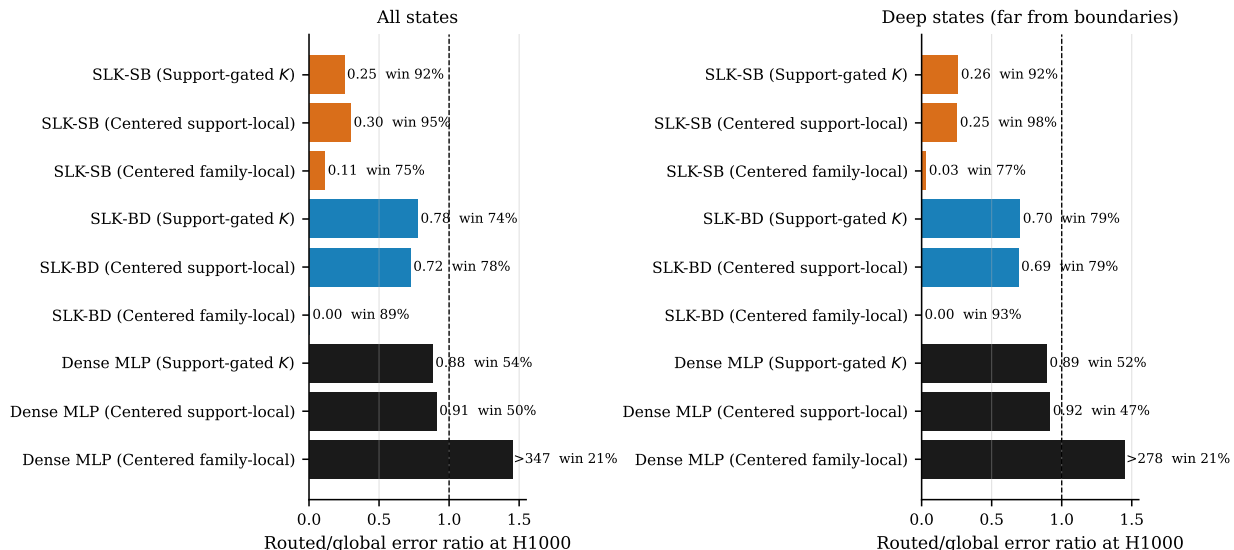


Figure 6: Routed/global $H1000$ error ratios on top-8 self-routed forecasting. Bars are clipped at 1.5; values to the right of the dashed line indicate that routing hurt. Numbers next to each bar are the IQM ratio and the win rate. The Dense MLP family-local route is unstable and is reported as the upper bound rather than the IQM.

are not directional at any horizon (6–10/15, no rejection). The crossover is therefore consistent with the multibasin reading: dense transitions track local detail better at moderate horizons, while a structured transition is more robust when small per-step errors accumulate over thousands of steps. Note that this benchmark has no basin labels; it therefore extends only the forecasting half of the chain.

6.7 Falsification: oracle, random, and latent-cluster controls

Three controls calibrate the routing claim. First, centered local-law fits conditioned on *oracle* basin labels beat the global K on 93–100% of deep evaluated rows for every model family, including the Dense MLP. This says local centred laws exist for many learned representations once a good partition is supplied; the sparse-latent contribution is specifically that the model itself supplies an inspectable, label-free partition that does this work. Second, random count-matched partitions and latent k -means clusters with the same number of classes do *not* reproduce the routed/global ratios in table 2: support-conditioned routing is sharper than partition-count or continuous-coefficient grouping alone. Third, on the local-Jacobian recovery diagnostic the picture is mixed: LISTA support families often beat random count-matched partitions near attractors, but the Dense MLP can reach lower absolute projected-Jacobian error because its representation is closer to the identity map. We therefore do not claim recovery of true Jacobian eigendirections; we claim support-conditioned local prediction and support-label agreement. The routing, refresh, and forecasting evidence together are tight; the geometric-recovery story remains a secondary, qualified diagnostic.

7 Related work

Koopman learning. Koopman operator theory represents nonlinear dynamics through a linear operator on observables (?Mezić, 2005). EDMD approximates this operator with a chosen dictionary

Table 3: Support refresh after controlled basin transfer (top-8 supports, ≥ 396 post-entry rollouts per cell). *Target dom.*: fraction of post-entry states whose refreshed exact support matches the target-basin canonical support. *Route-target*: fraction of post-entry steps that route through the target’s local class. *MSE ratio (IQM)*: cross-rollout interquartile mean of refreshed-current/frozen-source forecast MSE; range is the $[Q25, Q75]$ IQR for context. *K/12*: per-system one-sided paired Wilcoxon on $\log_{10}(\text{refreshed/frozen})$ clearing Holm-corrected $\alpha=0.05$, of 12 systems with ≥ 1 transfer pair.

Model	Period	Target dom.	Route-target	Fallback	MSE ratio: IQM $[Q25, Q75]$	K/12
LISTA-SB	1	0.83 (0.33)	0.86 (0.30)	0.14	8.5×10^{-3} $[1.0 \times 10^{-4}, 4.8 \times 10^{-4}]$	11/12
LISTA-SB	10	0.86 (0.24)	0.89 (0.19)	0.11	7.4×10^{-3} $[1.3 \times 10^{-4}, 6.5 \times 10^{-4}]$	11/12
LISTA-BD	1	0.07 (0.25)	0.74 (0.36)	0.26	0.58 $[0.07, 0.64]$	9/12
LISTA-BD	10	0.07 (0.25)	0.82 (0.26)	0.18	0.52 $[0.008, 0.18]$	11/12

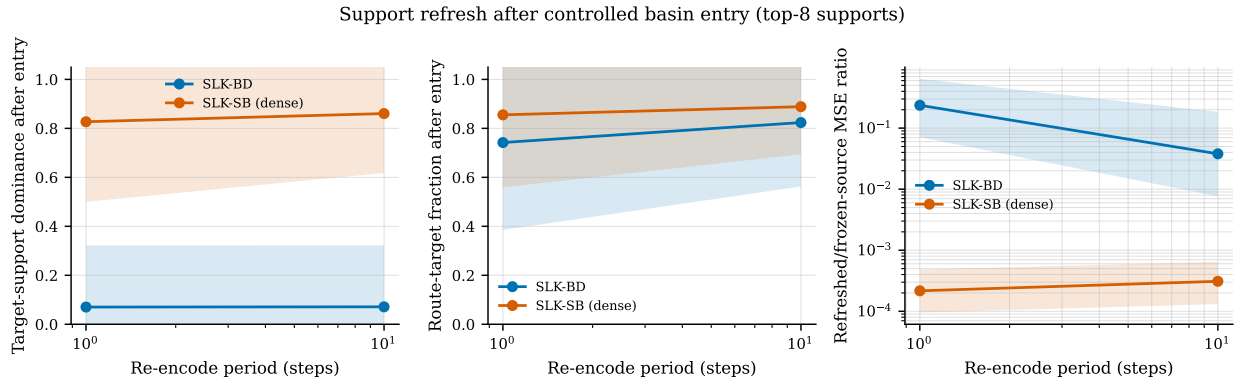


Figure 7: Post-entry behaviour for top-8 supports as a function of re-encode period. Left: target-support dominance. Centre: route-target fraction. Right: refreshed-current vs. frozen-source MSE ratio (log scale). Bands are cross-(system, seed) standard deviation (centre) or IQR (right).

(?). Neural Koopman methods learn the dictionary or invariant subspace from data (Takeishi et al., 2017; ?, Otto and Rowley, 2019). Consistent and course-corrected Koopman autoencoders improve rollout behavior through additional losses and correction mechanisms (Fathi et al., 2024). Our contribution is not a new claim that neural Koopman autoencoders can forecast. It is an empirical study of whether sparse latent supports carry information about the relevant basin or local dynamics.

Multibasin Koopman structure. Multiple invariant sets complicate naive interpretations of a single global finite-dimensional Koopman representation. Prior theory shows that global lifting and reconstruction depend on invariant-set structure, spectra, and symmetry (??). This motivates our use of support families and top- k supports rather than a universal exact-support claim.

Sparse coding and LISTA. Sparse coding is a classical representation principle in which signals are reconstructed from a small number of active atoms (?). LISTA learns fast amortized sparse inference (?). We use this sparse-inference bias inside a dynamical model and evaluate whether the resulting active atoms form support objects that agree with benchmark basin labels.

Table 4: Long-horizon Dysts forecasting. Each cell is the cross-system IQM of the per-system IQM-over-seeds (system standard deviation in parentheses), followed by $[K/15]$ – per-system one-sided paired Wilcoxon vs the Dense MLP baseline, Holm-corrected across the 15 systems at $\alpha=0.05$. At $n_{\text{seed}}=10$ the exact paired-Wilcoxon p -floor is $1/2^{10}\approx 10^{-3}$, just below Holm’s per-system threshold $\alpha/15\approx 3\times 10^{-3}$, but residual seed dispersion holds K to 0–1 in most cells; a one-sided binomial sign test on the per-system IQM of $\log_{10}$ -MSE deltas is reported in the body text and is the more sensitive within-system stat at this design. LISTA-D: dense-transition LISTA. LISTA-BD: block-diagonal LISTA at $\lambda_{\text{sp}} \in \{3\times 10^{-3}, 6\times 10^{-3}\}$. Sparse/Sparse MLP, BD/Dense MLP: matched standard-sampling MLP roots. Cross-system $\log_{10}$ standard deviation in [section E](#).

Model	H5000	H10000	H20000	H30000
LISTA-D	0.166 (0.57) [1/15]	0.957 (1.31) [1/15]	2.585 (1.92) [1/15]	3.223 (2.02) [1/15]
LISTA-BD (low sp)	0.640 (1.13) [0/15]	2.003 (1.62) [0/15]	2.816 (1.73) [0/15]	3.001 (1.62) [0/15]
LISTA-BD (high sp)	0.729 (1.07) [0/15]	1.694 (1.38) [0/15]	2.057 (1.60) [1/15]	2.235 (1.63) [1/15]
Sparse MLP	0.256 (0.71) [0/15]	1.168 (1.43) [1/15]	2.990 (1.60) [0/15]	3.786 (1.44) [0/15]
Sparse MLP, BD (low sp)	0.201 (0.71) [0/15]	1.171 (1.37) [0/15]	3.037 (1.78) [0/15]	3.669 (1.78) [0/15]
Sparse MLP, BD (high sp)	0.237 (0.82) [0/15]	1.211 (1.44) [0/15]	2.991 (1.80) [0/15]	3.504 (1.80) [0/15]
Dense MLP	0.259 (0.62)	1.433 (1.35)	3.035 (2.10)	3.618 (2.05)

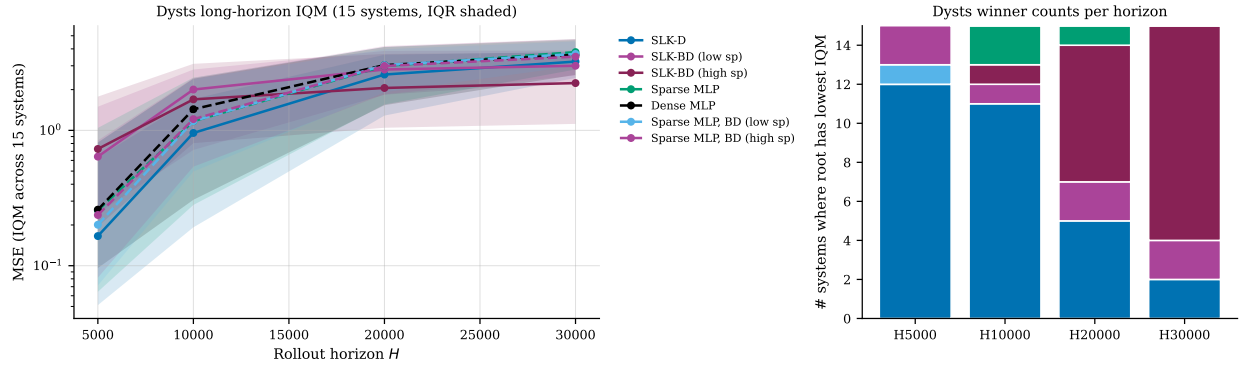


Figure 8: Dysts long-horizon forecasting. Left: IQM across the 15 Dysts systems with 25–75 percentile bands (log scale). Right: stacked counts of which root has the lowest per-system IQM at each horizon.

Switching and local linear models. Switching state-space models and local Koopman methods explicitly introduce modes or local operators (????). Our model differs because it does not train with known basin labels or a known basin count. The switching behavior, when it appears, is implicit in the learned sparse support.

8 Limitations and conclusion

The results support a qualified, falsifiable claim. Sparse-latent KAEs can learn support objects that agree with basin labels on states far from basin boundaries, and fixed-size top-8 supports can route useful local forecasts without oracle labels. Exact support uniqueness is not guaranteed globally. It is strongest away from separatrices and should not be used as a brittle thresholded deployment rule.

The evidence also separates sparsity from LISTA-specificity. Sparse MLP controls are strong, so the paper should not claim that LISTA is the only route to the observed behavior. The safer conclusion is that induced sparsity is a useful inductive bias for Koopman representations whose

active supports agree with basin labels, with LISTA providing one structured implementation of that bias.

Several limitations remain. The current true-geometry evidence is mixed, so we do not claim recovery of true Jacobian eigendirections. Controlled transfer between basins is positive for dense sparse-latent top-8 supports and for support families, but it is not uniformly positive for every transition structure or every exact support definition. The benchmark systems used for support-label comparisons are low-dimensional, and higher-dimensional systems may require additional structure or weaker support-family interpretations.

Overall, sparse support objects provide a useful bridge between forecasting and interpretability. They expose discrete structure related to basin membership and local dynamics inside a Koopman autoencoder, and the current evidence shows that this structure can agree with basin labels and route local predictive laws. The most robust paper claim is not that each basin has one exact support, but that induced sparse support objects can reveal and exploit local dynamical structure when evaluated with appropriate support definitions and falsification diagnostics.

A Support definitions and sensitivity

The absolute-threshold support S_{abs} marks a latent coordinate active when its magnitude exceeds 10^{-3} . It is the strict identifiability object used on the deep-state slice. The relative-threshold support S_{rel} marks a coordinate active when its magnitude is at least 10% of the largest latent magnitude for that state and is used for centered local-law diagnostics. The top- k support $S_{\text{top}k}$ selects the eight largest-magnitude latent coordinates. It is the deployment-like object because every state receives a route key without a magnitude threshold. Support families are produced by greedy Jaccard merging at threshold 0.5. In our results, S_{abs} saturates the conditional entropy $H(B | S)$ on the deep slice for all sparse rows and is therefore not a discriminator on its own; $H(S | B)$ and U_{exact} are the discriminative columns there. $S_{\text{top}8}$ is the only support definition for which the dense MLP control’s failure on routing cannot be explained by route-availability artifacts.

B Multibasin benchmark inventory

[Table 5](#) lists the 17 multibasin systems used in this paper, their basin counts available at evaluation, and the main-text display item that uses each system.

C Per-seed distributions

[Table 1](#) reports cross-system summaries of per-system IQM-over-seeds. The corresponding per-(system, seed) distributions are shown below. [Figure 9](#) plots per-(system, seed) histograms of forecasting MSE at every paper-facing horizon. [Figure 10](#) plots per-(system, seed) histograms of $H(B | S)$, $H(S | B)$, and U_{exact} on the deep-state slice. [Figure 11](#) plots the same per-seed distributions for the Dysts long-horizon benchmark. The distributions are heavy-tailed in log-MSE; we therefore report the cross-system $\log_{10}$ -IQR alongside the IQM in [table 1](#). The cross-system $\log_{10}$ -IQR is ≤ 1.97 for every row at every horizon (the highest values come from the rows whose worst-system seeds blow up at $H1000$, dominated by `multiwell_strong_transition`), and the per-system rank ordering of rows is stable across horizons.

Table 5: The 17 multibasin benchmark systems. Geometry types: *multiwell* = multistable potential systems; *gated* = gated local-linear systems with explicit transitions between regions; *polygonal* = arrangement of point attractors at vertices of a polygon; *checkerboard* = grid-based potential; *cascade* = depth-cascade variants. Basin counts are revealed only at evaluation.

System	Geometry	Basins	Used in
<code>multiwell_strong_transition</code>	multiwell	4	Tab. 1, Fig. 4
<code>gated_local_linear</code>	gated	4	Tab. 1, Tab. 2
<code>gated_transfer_linear</code>	gated	4	Tab. 1, Tab. 3
<code>arrested_spiral</code>	polygonal	4	Tab. 2
<code>cal_asymmetric_3</code>	polygonal	3	Tab. 1
<code>cal_high_cross_3</code>	polygonal	3	Tab. 1
<code>cal_hexagon_6</code>	polygonal	6	Tab. 1
<code>cal_octagon_8</code>	polygonal	8	Tab. 1
<code>cal_pentagon_5</code>	polygonal	5	Tab. 1
<code>cal_square_4</code>	polygonal	4	Tab. 1
<code>checkerboard_potential</code>	checkerboard	9	Tab. 1
<code>duffing_triple_well</code>	multiwell	3	Tab. 1, Fig. 4
<code>snic_multi</code>	polygonal	3	Tab. 1
<code>transition_routes_4</code>	gated	4	Tab. 1
<code>var_depth_gradient_4</code>	cascade	4	Tab. 1
<code>var_diamond_4</code>	polygonal	4	Tab. 1
<code>var_l_shape_5</code>	polygonal	5	Tab. 1

D Full per-system tables

Tables 6–8 render the per-(system, candidate) effect sizes and Holm-corrected Wilcoxon p -values that underlie the K/N counts in table 1 and the forest plot in figure 5. Each cell gives the per-system median paired $\log_{10}$ -MSE difference (candidate minus matched-sampling Dense MLP, no-shrink baseline) over the 10 training seeds and the corresponding Holm-corrected one-sided paired Wilcoxon p -value (Holm correction across the 17 systems within each horizon-and-candidate column). Bold entries indicate Holm-corrected $p < 0.05$ in favor of the candidate. Systems are sorted by the median across candidates of the per-system effect; the same ordering is used in all three tables for visual cross-reference.

The raw per-(system, seed) values, per-system bootstrap 95% CIs of the paired median, and the corresponding paired t -test p -values are released in the supplementary materials. The supplementary CSVs include per-system records, the row-level K/N summary, and the cross-system table used in the main text; the manifest lists the names and the source forecasting CSVs.

Per-system audit of the interpretability columns

Why the wrong-support-freeze denominator is 13 and not 17. The wrong-support-freeze diagnostic builds, for each system, a dictionary mapping each basin to its canonical support mask – the most-common exact support among the deep-slice states of that basin. The wrong-frozen rollout draws a mask from a basin different from the state’s own basin, so the test requires at least two distinct deep-slice canonical masks. Four of the 17 benchmark systems – `arrested_spiral` (5 basins), `cal_asymmetric_3` (3), `duffing_triple_well` (3), `var_l_shape_5` (5) – have multiple basins by design but a deep slice (top quartile by basin-depth margin) concentrated in a single basin: one basin’s attractor has a much larger margin to the second-nearest centroid than the others, so the global top-quartile filter selects only that basin’s states. This holds across all 7 trained model rows and all 9 support-threshold definitions, confirming it is a property of the system’s depth-margin geometry rather than of any encoder. The wrong-support-freeze test is undefined on those four

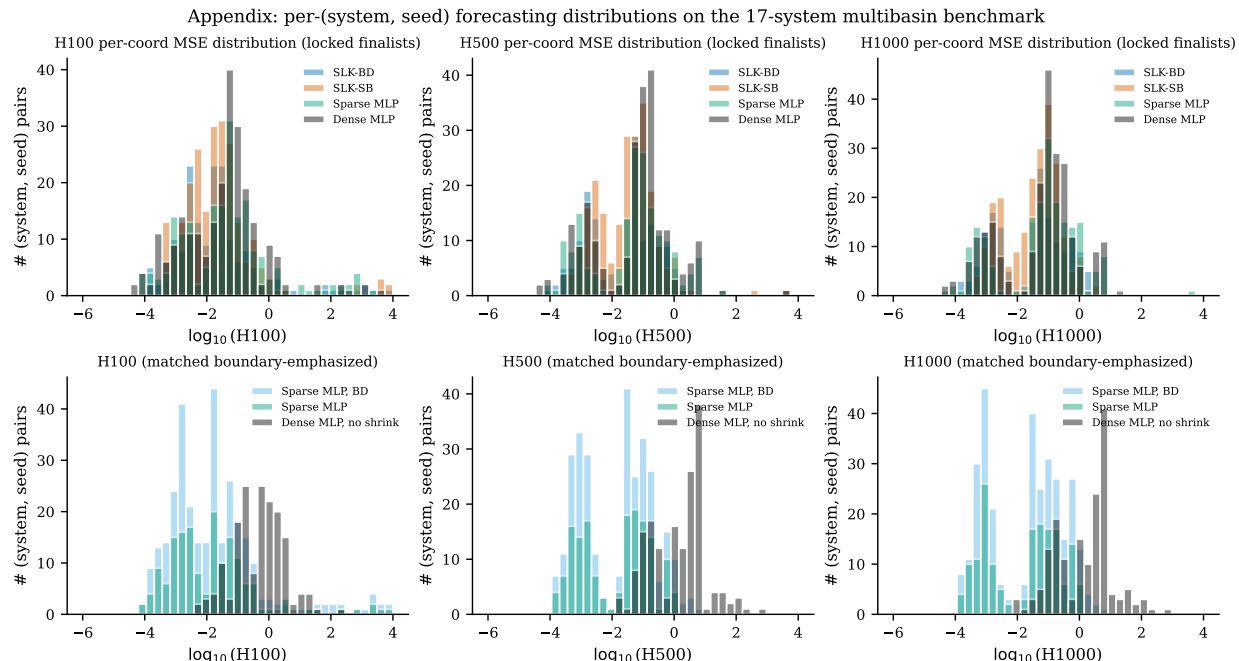


Figure 9: Per-(system, seed) forecasting MSE distributions on the multibasin benchmark. Each row is a horizon ($H100$, $H500$, $H1000$); the top row shows the LISTA finalists with the standard-sampling MLP controls (for completeness), and the bottom row shows the matched boundary-emphasized MLP controls used in the main-text comparison. Histograms are over the $\log_{10}$ raw MSE.

systems by construction, and the corresponding K/N denominator in [table 1](#) and [tables 11](#) and [12](#) is 13 instead of 17. The four omitted systems still appear in the per-system tables with “—” entries. A queued robustness re-evaluation under a per-basin top-quartile depth criterion – which admits each basin’s relative-deep states and is expected to restore 17/17 coverage on these four systems – is documented in `docs/PER_BASIN_DEEP_SLICE_PLAN.md`; its results will appear here as a separate appendix table once the re-eval completes, and the global-slice numbers in [table 1](#) and the per-system tables below are the source-of-truth in the meantime.

Tables [9–12](#) render the full per-(system, candidate) breakdown of the four interpretability columns of [table 1](#). Each cell gives the per-system median paired difference between the candidate and the matched-sampling Dense MLP no-shrink baseline (over the 10 training seeds), and the corresponding Holm-corrected one-sided paired Wilcoxon p -value (Holm correction across the 17 systems within each candidate column). Bold entries indicate Holm-corrected $p < 0.05$ in favor of the candidate. The alternative direction is “candidate < baseline” for $|S|$ and $H(B | F)$ (smaller is better) and “candidate > baseline” for the wrong-support-freeze ratios (larger is better – the support is more functionally meaningful when freezing the wrong one hurts more).

E Full per-system Dysts tables

The per-(system, seed) Dysts MSE values are released as `table3_dysts.csv` (cross-system summaries) and as `forecasting_rows.csv` for both Dysts evaluation packets. Cross-system $\log_{10}$ -MSE standard deviations range from 0.18 (LISTA-D at $H30000$) to 0.77 (LISTA-D at $H5000$); none exceed unity. The dense MLP control is consistent with these but has a wider per-system spread at

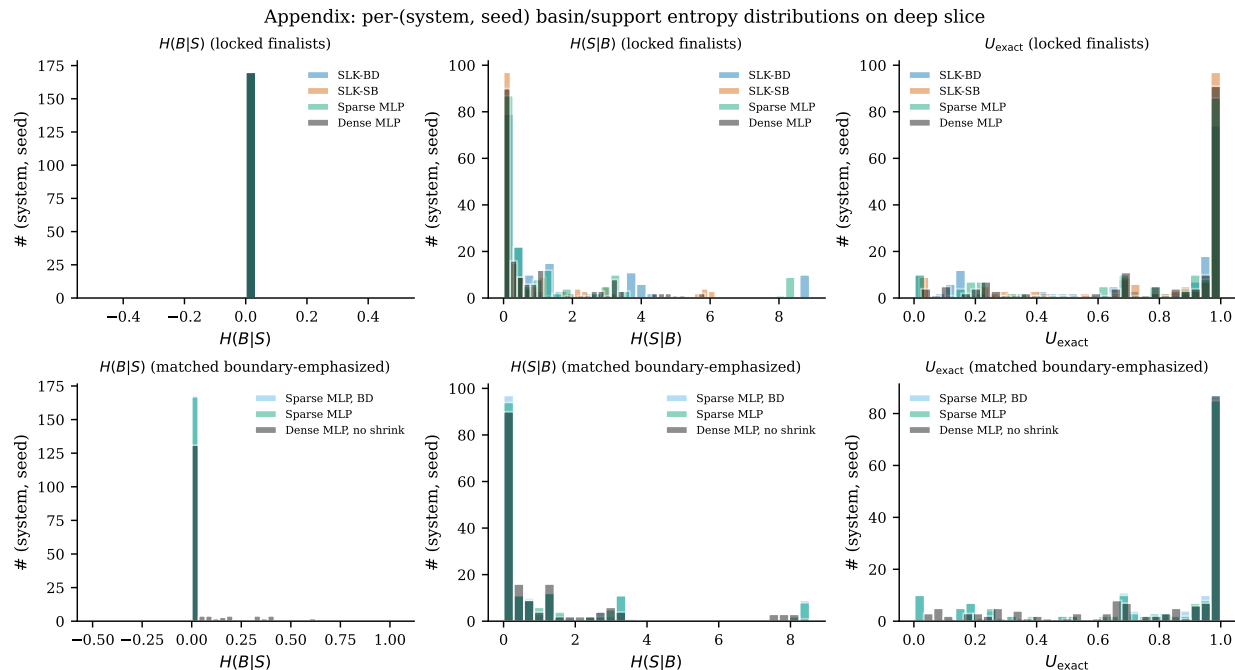


Figure 10: Per-(system, seed) histograms of basin/support agreement metrics on the deep-state slice (S_{abs}). Top row: LISTA finalists with standard-sampling MLP controls. Bottom row: matched boundary-emphasized MLP controls used in the main-text comparison. Lower is better for $H(B | S)$ and $H(S | B)$; higher is better for U_{exact} .

the longest horizons (figure 8 and figure 11).

F Falsification: oracle, random, and latent-cluster controls

The routing experiments include three falsification controls that we summarise here. (i) Oracle-conditioned A_{basin} fits beat the global K on 93–100% of evaluated deep rows for every model family, including the dense no-sparsity MLP. This says local centred laws exist for many representations once a good partition is supplied; the sparse-latent contribution is that the model itself supplies an inspectable, label-free partition that does this work. (ii) Random count-matched partitions and latent k -means clusters do not reproduce the routed/global ratios in table 2: support-conditioned routing improves on these controls by an order of magnitude on the deep slice for the sparse-latent rows. (iii) On the local-Jacobian recovery diagnostic, LISTA support families often beat random count-matched partitions near attractors but the dense no-sparsity MLP can have lower absolute projected-Jacobian error because its representation is closer to the identity map. We therefore do not claim recovery of true Jacobian eigendirections; we claim support-conditioned local prediction and support-label agreement.

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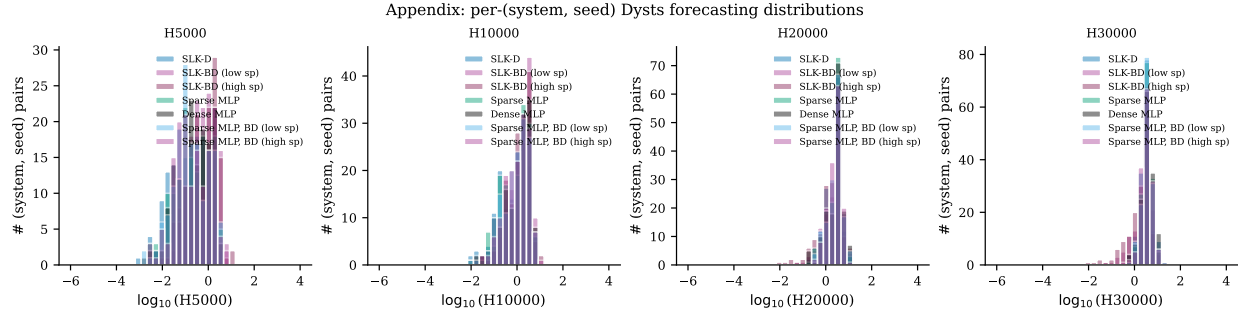


Figure 11: Per-(system, seed) forecasting MSE distributions on the Dysts long-horizon benchmark.

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Table 6: Per-system effect sizes and Holm-corrected Wilcoxon p -values at $H = 100$ on the 17-system multibasin benchmark, against the matched-sampling Dense MLP, no-shrink baseline. Δ is the per-system median paired $\log_{10}$ -MSE difference (candidate – baseline) over the 10 seeds; p_H is the one-sided paired Wilcoxon signed-rank p -value, Holm-corrected across the 17 systems within each candidate column. Bold Δ : Holm-corrected $p < 0.05$ in favor of the candidate.

System	LISTA-BD		LISTA-SB		Sparse MLP, BD		Sparse MLP	
	Δ	p_H	Δ	p_H	Δ	p_H	Δ	p_H
checkerboard_potential	−0.14	0.922	−0.75	0.131	+0.33	1.000	−0.75	0.193
snic_multi	−2.11	0.017	−1.62	0.017	−2.22	0.018	−2.22	0.017
gated_local_linear	−2.15	0.017	−2.10	0.017	−2.32	0.017	−2.50	0.017
var_lshape_5	−1.86	0.017	−1.72	0.017	−1.77	0.018	−2.04	0.017
cal_hexagon_6	−1.70	0.017	−1.83	0.017	−2.07	0.018	−1.96	0.018
arrested_spiral	−2.00	0.017	−1.17	0.029	−1.90	0.018	−1.83	0.018
cal_square_4	−1.68	0.017	−1.09	0.017	−1.85	0.017	−1.29	0.017
cal_pentagon_5	−1.33	0.017	−0.95	0.017	−1.56	0.017	−2.13	0.017
cal_octagon_8	−1.96	0.017	−1.89	0.017	−1.80	0.017	−1.67	0.017
cal_asymmetric_3	−1.82	0.017	−1.49	0.017	−1.70	0.018	−2.43	0.018
multiwell_strong_transition	+2.61	1.000	+1.52	0.990	+2.52	1.000	+2.34	0.999
transition_routes_4	−1.57	0.017	−2.20	0.017	−2.23	0.017	−2.41	0.017
var_diamond_4	−1.85	0.017	−1.71	0.017	−1.52	0.017	−1.80	0.018
var_depth_gradient_4	−1.54	0.017	−0.92	0.017	−1.31	0.018	−0.79	0.018
cal_high_cross_3	−1.23	0.017	−1.14	0.017	−1.16	0.017	−1.28	0.017
duffing_triple_well	−0.54	0.027	−0.60	0.017	−1.00	0.018	−0.98	0.018
gated_transfer_linear	−0.22	0.126	−0.28	0.017	−0.36	0.017	−0.25	0.126
$K/17$ Holm $p < 0.05$	$K=14 / N=17$		$K=15 / N=17$		$K=15 / N=17$		$K=14 / N=17$	

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Table 7: Per-system effect sizes and Holm-corrected Wilcoxon p -values at $H = 500$ on the 17-system multibasin benchmark. Format identical to table 6.

System	LISTA-BD		LISTA-SB		Sparse MLP, BD		Sparse MLP	
	Δ	p_H	Δ	p_H	Δ	p_H	Δ	p_H
checkerboard_potential	+6.46	1.000	+1.15	0.647	+2.83	1.000	−0.58	0.084
snic_multi	− 2.46	0.017	− 1.87	0.017	− 2.38	0.017	− 2.45	0.017
gated_local_linear	− 2.26	0.017	− 2.11	0.017	− 2.29	0.017	− 2.56	0.017
var_l_shape_5	− 1.89	0.017	− 2.21	0.017	− 2.42	0.017	− 2.76	0.017
cal_hexagon_6	− 1.94	0.017	− 2.13	0.017	− 2.45	0.017	− 2.33	0.017
arrested_spiral	− 2.43	0.017	+3.03	0.647	− 2.31	0.017	− 2.16	0.017
cal_square_4	− 1.92	0.017	− 1.32	0.017	− 2.26	0.017	− 1.49	0.017
cal_pentagon_5	− 1.71	0.017	− 1.27	0.017	− 1.98	0.017	− 2.73	0.017
cal_octagon_8	− 2.15	0.017	− 2.01	0.017	− 1.91	0.017	− 1.76	0.017
cal_asymmetric_3	− 1.98	0.017	− 1.92	0.017	− 2.27	0.017	− 3.29	0.017
multiwell_strong_transition	+23.90	1.000	+16.90	0.997	+24.96	1.000	+24.51	1.000
transition_routes_4	− 1.56	0.017	− 2.11	0.017	− 1.59	0.017	− 1.88	0.017
var_diamond_4	− 1.55	0.017	− 1.26	0.017	− 1.45	0.017	− 2.06	0.017
var_depth_gradient_4	− 1.81	0.017	− 1.05	0.017	− 1.56	0.017	− 0.92	0.017
cal_high_cross_3	− 1.17	0.017	− 1.14	0.017	− 1.15	0.017	− 1.32	0.017
duffing_triple_well	− 0.88	0.017	− 0.83	0.017	− 1.32	0.017	− 1.24	0.017
gated_transfer_linear	+1.85	0.158	− 0.16	0.017	− 0.16	0.017	− 0.19	0.017
$K/17$ Holm $p < 0.05$	$K=14 / N=17$		$K=14 / N=17$		$K=15 / N=17$		$K=15 / N=17$	

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Table 8: Per-system effect sizes and Holm-corrected Wilcoxon p -values at $H = 1000$ on the 17-system multibasin benchmark. This is the numerical version of the forest plot in figure 5. Format identical to table 6.

System	LISTA-BD		LISTA-SB		Sparse MLP, BD		Sparse MLP	
	Δ	p_H	Δ	p_H	Δ	p_H	Δ	p_H
checkerboard_potential	-2.78	0.216	-2.55	0.017	-2.90	0.017	-2.61	0.017
snic_multi	-2.37	0.017	-1.88	0.017	-2.42	0.017	-2.39	0.017
gated_local_linear	-2.23	0.017	-2.21	0.017	-2.38	0.017	-2.70	0.017
var_l_shape_5	-1.89	0.017	-2.07	0.017	-2.45	0.017	-2.55	0.017
cal_hexagon_6	-1.92	0.017	-2.01	0.017	-2.14	0.017	-2.14	0.017
arrested_spiral	-2.14	0.017	-1.81	0.129	-2.08	0.017	-2.05	0.017
cal_square_4	-2.06	0.017	-1.28	0.017	-2.26	0.017	-2.04	0.017
cal_pentagon_5	-1.70	0.017	-1.31	0.017	-1.98	0.017	-2.83	0.017
cal_octagon_8	-2.02	0.017	-1.79	0.017	-1.87	0.017	-1.65	0.017
cal_asymmetric_3	-1.75	0.017	-1.48	0.017	-1.77	0.017	-3.60	0.017
multiwell_strong_transition	-1.63	0.084	-2.42	0.500	-1.70	0.042	-1.75	0.042
transition_routes_4	-1.35	0.017	-1.72	0.017	-1.55	0.017	-1.63	0.017
var_diamond_4	-1.57	0.017	-1.32	0.017	-1.57	0.017	-2.00	0.017
var_depth_gradient_4	-2.00	0.017	-1.05	0.017	-2.27	0.017	-0.55	0.017
cal_high_cross_3	-1.19	0.017	-1.24	0.017	-1.07	0.017	-1.38	0.017
duffing_triple_well	-0.81	0.017	-0.75	0.017	-1.21	0.017	-1.09	0.017
gated_transfer_linear	-0.15	0.017	-0.13	0.017	-0.16	0.017	-0.17	0.017
$K/17$ Holm $p < 0.05$	$K=15 / N=17$		$K=15 / N=17$		$K=17 / N=17$		$K=17 / N=17$	

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Table 9: Per-system effect sizes and Holm-corrected Wilcoxon p -values for $|S|$ (mean active-coordinate count on the deep slice) on the 17-system multibasin benchmark. Alternative: candidate $|S| <$ baseline $|S|$. Lower (more negative) is better for the candidate. Latent dimension is $d_z=64$ for LISTA-SB and $d_z=256$ for LISTA-BD and the MLP rows.

System	LISTA-BD		LISTA-SB		Sparse MLP, BD		Sparse MLP	
	Δ	p_H	Δ	p_H	Δ	p_H	Δ	p_H
snic_multi	-174.91	0.017	-233.11	0.017	-171.26	0.017	-171.16	0.017
gated_local_linear	-167.78	0.017	-229.59	0.017	-160.43	0.017	-160.28	0.017
cal_asymmetric_3	-158.00	0.017	-229.02	0.017	-159.00	0.017	-162.00	0.017
gated_transfer_linear	-154.26	0.017	-227.24	0.017	-157.98	0.017	-158.41	0.017
transition_routes_4	-162.97	0.017	-228.84	0.017	-152.28	0.017	-152.68	0.017
duffing_triple_well	-163.37	0.017	-230.04	0.017	-149.16	0.017	-149.16	0.017
var_diamond_4	-147.30	0.017	-226.48	0.017	-155.50	0.017	-155.61	0.017
arrested_spiral	-149.91	0.017	-225.92	0.017	-154.04	0.017	-154.19	0.017
cal_pentagon_5	-156.14	0.017	-225.99	0.017	-142.92	0.017	-142.86	0.017
multiwell_strong_transition	-137.88	0.017	-225.31	0.017	-149.39	0.017	-149.34	0.017
cal_hexagon_6	-153.12	0.017	-227.34	0.017	-143.44	0.017	-143.44	0.017
var_l_shape_5	-151.37	0.017	-227.92	0.017	-143.60	0.017	-143.42	0.017
cal_square_4	-152.42	0.017	-225.23	0.017	-141.23	0.017	-141.23	0.017
var_depth_gradient_4	-150.24	0.017	-225.29	0.017	-140.58	0.017	-140.24	0.017
cal_octagon_8	-146.49	0.017	-226.48	0.017	-143.17	0.017	-143.17	0.017
checkerboard_potential	-143.32	0.017	-221.69	0.017	-138.58	0.017	-138.58	0.017
cal_high_cross_3	-144.04	0.017	-226.00	0.017	-134.51	0.017	-135.50	0.017
$K/17$ Holm $p < 0.05$	$K=17 / N=17$		$K=17 / N=17$		$K=17 / N=17$		$K=17 / N=17$	

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Table 10: Per-system effect sizes and Holm-corrected Wilcoxon p -values for $H(B | F)$ (basin uncertainty given the support *family*, deep slice). Alternative: candidate $H(B | F) < \text{baseline}$. Lower is better for the candidate.

System	LISTA-BD		LISTA-SB		Sparse MLP, BD		Sparse MLP	
	Δ	p_H	Δ	p_H	Δ	p_H	Δ	p_H
transition_routes_4	-1.38	0.017	-1.38	0.017	-1.38	0.017	-1.38	0.017
cal_square_4	-1.37	0.017	-0.86	0.017	-1.37	0.017	-1.37	0.017
var_diamond_4	-1.37	0.017	-1.22	0.017	-1.37	0.017	-1.37	0.020
multiwell_strong_transition	-1.26	0.017	-1.23	0.017	-1.29	0.017	-1.29	0.017
cal_octagon_8	-1.34	0.017	-1.09	0.017	-1.09	0.017	-1.09	0.017
snic_multi	-1.09	0.017	-1.09	0.017	-1.09	0.018	-1.09	0.017
gated_local_linear	-1.07	0.017	-1.07	0.017	-1.07	0.017	-1.07	0.017
gated_transfer_linear	-1.06	0.017	-1.06	0.017	-1.06	0.017	-1.06	0.017
cal_hexagon_6	-0.92	0.017	-0.66	0.017	-0.77	0.018	-0.79	0.020
cal_pentagon_5	-0.64	0.023	+0.00	0.375	-0.64	0.023	-0.64	0.027
var_depth_gradient_4	-0.62	0.017	-0.31	0.027	-0.62	0.018	-0.62	0.027
checkerboard_potential	-1.24	0.017	-0.54	0.017	-0.14	0.017	-0.13	0.020
cal_high_cross_3	-0.01	0.023	+0.00	0.375	-0.01	0.078	-0.01	0.039
cal_asymmetric_3	+0.00	1.000	+0.00	1.000	+0.00	1.000	+0.00	1.000
duffing_triple_well	+0.00	1.000	+0.00	1.000	+0.00	1.000	+0.00	1.000
arrested_spiral	+0.00	1.000	+0.00	1.000	+0.00	1.000	+0.00	1.000
var_l_shape_5	+0.00	1.000	+0.00	1.000	+0.00	1.000	+0.00	1.000
$K/17$ Holm $p < 0.05$	$K=13 / N=17$		$K=11 / N=17$		$K=12 / N=17$		$K=13 / N=17$	

Table 11: Per-system effect sizes and Holm-corrected Wilcoxon p -values for the support-freeze wrong-over-base MSE ratio at $h=1$ (deep slice). Alternative: candidate ratio $> \text{baseline}$. Higher is better for the candidate (wrong-support freezing hurts the prediction more, indicating the model genuinely uses the support).

System	LISTA-BD		LISTA-SB		Sparse MLP, BD		Sparse MLP	
	Δ	p_H	Δ	p_H	Δ	p_H	Δ	p_H
multiwell_strong_transition	+497.20	0.013	+828.73	0.013	+1134.12	0.013	+1477.16	0.013
var_depth_gradient_4	+4851.48	0.013	+802.44	0.013	+1086.62	0.013	+1121.45	0.013
cal_pentagon_5	+4857.91	0.013	+636.10	0.013	+1679.06	0.013	+1388.96	0.013
cal_high_cross_3	+1359.69	0.013	+1082.86	0.013	+2653.52	0.013	+3301.41	0.013
checkerboard_potential	+1161.51	0.013	+731.70	0.013	+3083.50	0.013	+3577.67	0.013
cal_octagon_8	+6579.32	0.013	+3159.16	0.013	+3563.95	0.013	+3563.95	0.013
cal_square_4	+15284.66	0.013	+2200.14	0.013	+5580.87	0.013	+5249.74	0.013
var_diamond_4	+5602.16	0.013	+4745.27	0.013	+7947.22	0.013	+10718.65	0.013
gated_transfer_linear	+6274.01	0.013	+3317.50	0.013	+8587.81	0.013	+7348.76	0.013
cal_hexagon_6	+7829.62	0.013	+5473.99	0.013	+12232.14	0.013	+11641.91	0.013
transition_routes_4	+108283.56	0.013	+21622.27	0.013	+35090.81	0.013	+26428.24	0.013
snic_multi	+20622.20	0.013	+12107.59	0.013	+125008.24	0.013	+407479.13	0.013
gated_local_linear	+74134.78	0.013	+65733.62	0.013	+425426.24	0.013	+350747.37	0.013
$K/17$ Holm $p < 0.05$	$K=13 / N=13$		$K=13 / N=13$		$K=13 / N=13$		$K=13 / N=13$	

Table 12: Per-system effect sizes and Holm-corrected Wilcoxon p -values for the support-freeze wrong-over-base MSE ratio at $h=20$. Alternative: candidate $>$ baseline. Higher is better.

System	LISTA-BD		LISTA-SB		Sparse MLP, BD		Sparse MLP	
	Δ	p_H	Δ	p_H	Δ	p_H	Δ	p_H
multiwell_strong_transition	-0.61	0.652	-0.43	0.770	-0.64	0.968	-0.71	0.993
cal_octagon_8	+4.51	0.013	+3.76	0.021	+11.49	0.013	+5.76	0.013
cal_pentagon_5	+11.60	0.013	+3.19	0.048	+7.42	0.013	+6.30	0.013
cal_square_4	+7.09	0.013	+0.56	0.387	+8.20	0.013	+9.21	0.013
var_depth_gradient_4	+1.83	0.158	-0.17	0.770	+18.37	0.013	+19.03	0.021
cal_high_cross_3	+0.32	0.432	+0.91	0.387	+22.91	0.014	+36.89	0.021
checkerboard_potential	+3.99	0.020	+11.00	0.026	+27.07	0.013	+41.31	0.013
gated_transfer_linear	+25.19	0.013	+8.56	0.122	+43.78	0.013	+20.78	0.013
cal_hexagon_6	+15.98	0.013	+0.76	0.117	+45.83	0.013	+45.83	0.013
var_diamond_4	+25.89	0.013	+4.07	0.026	+69.31	0.013	+55.66	0.013
transition_routes_4	+139.16	0.013	+17.63	0.021	+217.00	0.013	+149.91	0.013
snic_multi	+390.23	0.013	+45.01	0.013	+473.34	0.013	+627.79	0.013
gated_local_linear	+834.09	0.013	+1763.95	0.013	+6860.15	0.013	+5992.93	0.013
$K/17$ Holm $p < 0.05$	$K=10 / N=13$		$K=7 / N=13$		$K=12 / N=13$		$K=12 / N=13$	